

Chapter 1

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Exercise 1. A nonempty family of sets $\mathcal{R} \subset \mathcal{P}(X)$ is called a **ring** if it is closed under finite unions and differences (i.e., if $E_1, \dots, E_n \in \mathcal{R}$, then $\cup_1^n E_j \in \mathcal{R}$, and if $E, F \in \mathcal{R}$, then $E \setminus F \in \mathcal{R}$). A ring that is closed under countable unions is called a **σ -ring**.

- a. Rings (resp. σ -rings) are closed under finite (resp. countable) intersections.
- b. If $\mathcal{R}$ is a ring (resp. σ -ring), then $\mathcal{R}$ is an algebra (resp. σ -algebra) iff $X \in \mathcal{R}$.
- c. If $\mathcal{R}$ is a σ -ring, then $\{E \subset X : E \in \mathcal{R} \text{ or } E^c \in \mathcal{R}\}$ is a σ -algebra.
- d. If $\mathcal{R}$ is a σ -ring, then $\{E \subset X : E \cap F \in \mathcal{R} \text{ for all } F \in \mathcal{R}\}$ is a σ -algebra.

Proof.

- a. If $\mathcal{R}$ is a σ -ring and $\{E_n\}_{n \in \mathbb{N}} \subset \mathcal{R}$, then

$$\bigcap_{n \in \mathbb{N}} E_n = \left(\bigcup_{n \in \mathbb{N}} E_n \right) \setminus \left(\bigcup \{E_a \setminus E_b : a, b \in \mathbb{N}\} \right) \quad (1)$$

Considering the left and right components of the difference in the right hand side of (1), the left is clearly a countable union of members of $\mathcal{R}$, and the right can be seen to be a countable union by identifying each $E_a \setminus E_b$ with $\frac{a}{b}$. Thus the right hand side of (1) is a difference of countable unions of members of $\mathcal{R}$, so it is in $\mathcal{R}$.

If $\mathcal{R}$ is a ring, the argument above applies by making $\{E_n\}$ a finite set.

- b. In either case, if $\mathcal{R}$ is a ring (resp. σ -ring) and an algebra (resp. σ -algebra), then given $E \in \mathcal{R}$, $X = E \cup E^c \in \mathcal{R}$.

Conversely, if $\mathcal{R}$ is a ring (resp. σ -ring) and $X \in \mathcal{R}$, then given $E \in \mathcal{R}$, $E^c = X \setminus E \in \mathcal{R}$, so $\mathcal{R}$ is closed under complements. $\mathcal{R}$ is already closed under finite (resp. countable) unions, so $\mathcal{R}$ is an algebra.

- c. Let $A = \{E \subset X : E \in \mathcal{R} \text{ or } E^c \in \mathcal{R}\}$. Given $E \in A$, by definition $E^c \in A$ since $(E^c)^c = E \in A$.

Let $\{E_n\}_{n \in \mathbb{N}} \subset A$, and express this set as the union of two sets, namely

$$\{E_n\}_{n \in \mathbb{N}} = \{F_j\}_{j \in J} \cup \{G_k\}_{k \in K}$$

where $\{F_j\}_{j \in J} \subset A$ is the collection of all sets in A who are members of $\mathcal{R}$ and $\{G_k\}_{k \in K} \subset A$ is the collection of sets in A who are not themselves in $\mathcal{R}$, but whose complements are. Then

$$\begin{aligned} \bigcup_{n \in \mathbb{N}} E_n &= \bigcup_{j \in J} F_j \cup \bigcup_{k \in K} G_k \\ &= \bigcup_{j \in J} F_j \cup \left(\bigcap_{k \in K} G_k^c \right)^c \end{aligned} \quad (1)$$

Since each F_j is in $\mathcal{R}$, $\bigcup_{j \in J} F_j \in \mathcal{R} \subset A$. Also, we know from (a) that σ -rings are closed under countable intersections, so since $G_k^c \in \mathcal{R}$ for each G_k , we have $\bigcap_{k \in K} G_k^c \in \mathcal{R}$. Its complement is also in $\mathcal{R}$ since $\mathcal{R}$ is closed under complementation. Thus (2) is a union of two members of $\mathcal{R}$, and is therefore in $\mathcal{R} \subset A$.

- d. Let $A = \{E \subset X : E \cap F \in \mathcal{R} \text{ for all } F \in \mathcal{R}\}$. Note that $\mathcal{R} \subset A$ since $\mathcal{R}$ is closed under finite intersections. Given any $E, F \in \mathcal{R}$, we have

$$E^c \cap F = (X \setminus E) \cap F = F \setminus E$$

which is a difference in $\mathcal{R}$ and is thus in $\mathcal{R} \subset A$.

Given $\{E_j\}_{j \in \mathbb{N}} \subset A$ and arbitrary $F \in A$,

$$\left(\bigcup_{j \in \mathbb{N}} E_j \right) \cap F = \bigcup_{j \in \mathbb{N}} (E_j \cap F),$$

which is a countable union of elements in $\mathcal{R}$. Thus since the choice of F was arbitrary, $\bigcup \{E_j\} \in A$.

□

Exercise 2. Complete the proof of Proposition 1.2.

Proof. The construction of open intervals from half-open intervals and rays is standard. □

Exercise 3. Let $\mathcal{M}$ be an infinite σ -algebra.

- a. $\mathcal{M}$ contains an infinite sequence of disjoint nonempty sets.
- b. $\text{card}(\mathcal{M}) \geq c$.

Proof.

- a. Start with $E_1, E_2 \in \mathcal{M}$, both not X . Take $E_3 \in \mathcal{M}$ not a subset of $E_1 \cup E_2$ and not X . Continue choosing E_i such that $E_i \neq X$ and $E_i \not\subset \bigcup_{j=1}^{i-1} E_j$ for all $i \in \mathbb{Z}^+$. Assume that for some i , such a choice is impossible. Then since $\mathcal{M}$ is infinite, there must be infinitely many subsets of $\bigcup_{j=1}^{i-1} E_j$ in $\mathcal{M}$. Each of these subsets has a complement which is not a subset of $E_1 \cup E_2$, contradicting the assumption of impossibility.

Finally, the sequence $\{F_i\}$, where $F_i = E_i \setminus \bigcup_{j=1}^{i-1} E_j$ is a sequence of disjoint members of $\mathcal{M}$ as desired.

- b. Choose a sequence $\{E_j\}_1^\infty$ of disjoint nonempty sets in $\mathcal{M}$ (allowable by part (a) of this exercise). Define $f : \mathcal{P}(\mathbb{N}) \rightarrow \mathcal{M}$ by $f : S \mapsto \bigcup_{k \in S} E_k$. This is well-defined since $f(S)$ is a countable union of members of $\mathcal{M}$, so $f(S) \in \mathcal{M}$. Also, f is an injection since all the E_j are disjoint. This brings us to the conclusion that since $\text{im } f \subset \mathcal{M}$,

$$\text{card } \mathcal{M} \geq \text{card}(\text{im } f) = \text{card}(\mathcal{P}(\mathbb{N})) = c.$$

□

Exercise 4. An algebra $\mathcal{A}$ is a σ -algebra iff $\mathcal{A}$ is closed under countable increasing unions (i.e., if $\{E_j\}_1^\infty \subset \mathcal{A}$ and $E_1 \subset E_2 \subset \dots$, then $\bigcup_1^\infty E_j \in \mathcal{A}$).

Proof. If $\mathcal{A}$ is a σ -algebra, then $\mathcal{A}$ is closed under countable unions in general, so it is certainly closed under countable increasing unions in particular. On the other hand, if $\mathcal{A}$ is an algebra which is closed under countable increasing unions, then take any $\{E_j\}_1^\infty \subset \mathcal{A}$. Consider that

$$\bigcup_{j=1}^\infty E_j = \bigcup_{j=1}^\infty \bigcup_{k=1}^j E_k. \quad (1)$$

Clearly, for any $j \in \mathbb{N}$, $\left(\bigcup_{k=1}^j E_k\right) \subset \left(\bigcup_{k=1}^{j+1} E_k\right)$, and each $\bigcup_{k=1}^j E_k$ is a finite union of elements of the algebra $\mathcal{A}$, and thus is in $\mathcal{A}$. Then the right hand side of (1) is a countable increasing union of members of $\mathcal{A}$. Thus $(1) \in \mathcal{A}$. □

Exercise 5. If $\mathcal{M}$ is the σ -algebra generated by $\mathcal{E}$, then $\mathcal{M}$ is the union of the σ -algebras generated by $\mathcal{F}$ as $\mathcal{F}$ ranges over all countable subsets of $\mathcal{E}$.

Proof. Let F be the set of all countable subsets of $\mathcal{E}$. By Lemma 1.1, since $\mathcal{F} \subset \mathcal{E} \subset \mathcal{M}(\mathcal{E})$ for each $\mathcal{F} \in F$, $\mathcal{M}(\mathcal{F}) \subset \mathcal{M}(\mathcal{E})$. Their union, then, is also a subset of $\mathcal{E}$.

We need to show, then, that $\mathcal{M}(\mathcal{E}) \subset \bigcup_{\mathcal{F} \in F} \mathcal{M}(\mathcal{F})$. If we can show that $\bigcup_{\mathcal{F} \in F} \mathcal{M}(\mathcal{F}) = \mathcal{M}(\bigcup_{\mathcal{F} \in F} \mathcal{F})$, then we will be done because $\mathcal{E} \subset \bigcup F$, so $\mathcal{M}(\mathcal{E}) \subset \mathcal{M}(\bigcup F)$. That is where we now turn our focus.

Clearly $\mathcal{M}(\bigcup F) \subset \bigcup_{\mathcal{F} \in F} \mathcal{M}(\mathcal{F})$ since $\bigcup F \subset \bigcup_{\mathcal{F} \in F} \mathcal{M}(\mathcal{F})$, so we need to prove the other direction. Given $E \in \bigcup \mathcal{M}(\mathcal{F})$, fix some $\mathcal{F} \in F$ for which $E \in \mathcal{M}(\mathcal{F})$. Then since $\mathcal{F} \subset \bigcup F$, $\mathcal{M}(\mathcal{F}) \subset \mathcal{M}(\bigcup F)$, so $E \in \mathcal{M}(\bigcup F)$. As observed above, this completes our proof. □

Exercise 6. Complete the proof of Theorem 1.9.

Proof. First we verify that $\bar{\mu}$ is a complete measure on $\bar{\mathcal{M}}$. Since we can write $\emptyset = \emptyset \cup \emptyset$ where $\emptyset \in \mathcal{M}, \mathcal{N}$, we have $\bar{\mu} = \mu(\emptyset) = 0$. Given a disjoint collection $\{E_j \cup F_j\}_{j \in \mathbb{N}} \subset \bar{\mathcal{M}}$ where each $E_j \in \mathcal{M}$ and each $F_j \subset N_j \in \mathcal{N}$, we have

$$\begin{aligned} \bar{\mu} \left(\bigcup_1^\infty (E_j \cup F_j) \right) &= \bar{\mu} \left(\bigcup_1^\infty E_j \cup \bigcup_1^\infty F_j \right) \\ &= \mu \left(\bigcup_1^\infty E_j \right) \\ &= \sum_1^\infty \mu(E_j) \\ &= \sum_1^\infty \bar{\mu}(E_j \cup F_j) \end{aligned}$$

where the second equality holds because $\bigcup_1^\infty E_j \in \mathcal{M}$ and $\bigcup_1^\infty F_j \in \mathcal{N}$ since both $\mathcal{M}$ and $\mathcal{N}$ are closed under countable unions. Of course, since $\bar{\mu}$ is a measure and its domain is complete, $\bar{\mu}$ is a complete measure.

Now we turn our attention to the uniqueness of $\bar{\mu}$ as an extension of μ over $\bar{\mathcal{M}}$. Assume that f is a measure over $\bar{\mathcal{M}}$ which extends μ . Then given $M \cup F \in \bar{\mathcal{M}}$ where $M \in \mathcal{M}$ and $F \subset N \in \mathcal{N}$, by monotonicity we have that $f(F) \leq f(N) = 0$, so $f(F) = 0$. We have

$$f(M) \leq f(M \cup F) \leq f(M) + f(F) = f(M).$$

Thus $f(M \cup F) = f(M) = \mu(M) = \bar{\mu}(M \cup F)$. □

Exercise 7. If $\mu_1, \dots, \mu_n$ are measures on $(X, \mathcal{M})$ and $a_1, \dots, a_n \in [0, \infty)$, then $\sum_1^n a_j \mu_j$ is a measure on $(X, \mathcal{M})$.

Proof. Clearly $\sum_1^\infty a_j \mu_j(\emptyset) = 0$. Given a disjoint collection $\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{M}$, we have

$$\begin{aligned} \sum_{k=1}^n \left(a_k \mu_k \left(\bigcup_{j=1}^\infty E_j \right) \right) &= \sum_{k=1}^n \left(a_k \sum_{j=1}^\infty \mu_k(E_j) \right) \\ &= \sum_{k=1}^n \sum_{j=1}^\infty a_k \mu_k(E_j) \\ &= \sum_{j=1}^\infty \sum_{k=1}^n a_k \mu_k(E_j). \end{aligned}$$

□

Exercise 8. If $(X, \mathcal{M}, \mu)$ is a measure space and $\{E_j\}_{j=1}^\infty \subset \mathcal{M}$, then $\mu(\liminf E_j) \leq \liminf \mu(E_j)$. Also, $\mu(\limsup E_j) \geq \limsup \mu(E_j)$ provided that $\mu(\bigcup_1^\infty E_j) < \infty$.

Proof. Observe that

$$\begin{aligned} \mu(\liminf E_j) &= \mu\left(\bigcup_{k=1}^\infty \bigcap_{j=k}^\infty E_j\right) \\ &= \lim_{k \rightarrow \infty} \mu\left(\bigcap_{j=k}^\infty E_j\right) \\ &\leq \lim_{k \rightarrow \infty} \mu(E_k) \\ &\dots \text{UNFINISHED} \\ &= \sup_{k \geq 1} \inf_{j \geq k} \mu(E_j) \\ &= \liminf \mu(E_j) \end{aligned}$$

□

Exercise 9. If $(X, \mathcal{M}, \mu)$ is a measure space and $E, F \in \mathcal{M}$, then $\mu(E) + \mu(F) = \mu(E \cup F) + \mu(E \cap F)$.

Proof.

$$\begin{aligned} \mu(E \cup F) &= \mu((E \setminus F) \cup (E \cap F) \cup (F \setminus E)) \\ &= \mu(E \setminus F) + \mu(E \cap F) + \mu(F \setminus E) \\ &= \mu(E \setminus F) + \mu(E \cap F) - \mu(E \cap F) + \mu(E \cap F) + \mu(F \setminus E) \\ &= \mu((E \setminus F) \cup (E \cap F)) - \mu(E \cap F) + \mu((E \cap F) \cup (F \setminus E)) \\ &= \mu(E) - \mu(E \cap F) + \mu(F). \end{aligned}$$

Thus, $\mu(E \cup F) + \mu(E \cap F) = \mu(E) + \mu(F)$ as desired. □

Exercise 10. Given a measure space $(X, \mathcal{M}, \mu)$ and $E \in \mathcal{M}$, define $\mu_E(A) = \mu(A \cap E)$ for $A \in \mathcal{M}$. Then μ_E is a measure.

Proof. First off, $\mu_E(\emptyset) = \mu(\emptyset \cap E) = \mu(\emptyset) = 0$. Now suppose that we have some

$\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{M}$ a disjoint collection. Then

$$\begin{aligned}
\mu_E \left(\bigcup_{j=1}^{\infty} E_j \right) &= \mu \left(\left(\bigcup_{j=1}^{\infty} E_j \right) \cap E \right) \\
&= \mu \left(\bigcup_{j=1}^{\infty} E_j \cap E \right) \\
&= \sum_{j=1}^{\infty} \mu(E_j \cap E) \\
&= \sum_{j=1}^{\infty} \mu_E(E_j).
\end{aligned}$$

□

Exercise 11. A finitely additive measure μ is a measure iff it is continuous from below as in Theorem 1.8c. If $\mu(X) < \infty$, μ is a measure iff it is continuous from above as in Theorem 1.8d.

Proof. For both assertions, if μ is a measure, then clearly it is continuous from above and below so we restrict ourselves to proving the other direction in either case. Also, $\mu(\emptyset) = 0$ since μ is a finitely additive measure in both cases, so we will only prove countable additivity.

Given that μ is a finitely additive measure and it is continuous from below, consider a disjoint collection $\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{M}$. Consider the sequence defined by $F_j = \bigcup_{k=1}^j E_k$, which is clearly an ascending chain. Then

$$\begin{aligned}
\mu \left(\bigcup_{j=1}^{\infty} E_j \right) &= \mu \left(\bigcup_{j=1}^{\infty} F_j \right) \\
&= \lim_{j \rightarrow \infty} \mu(F_j) \\
&= \lim_{j \rightarrow \infty} \mu \left(\bigcup_{k=1}^j E_k \right) \\
&= \lim_{j \rightarrow \infty} \sum_{k=1}^j \mu(E_k) \\
&= \sum_{j=1}^{\infty} \mu(E_j).
\end{aligned}$$

Similarly, assuming that $\mu(X) < \infty$ and μ is continuous from above, define

$\{F_j\}$ as above. Observe that $\{F_j^C\}$ is a descending chain, so

$$\begin{aligned}
\mu\left(\bigcup_{j=1}^{\infty} E_j\right) &= \mu\left(\bigcup_{j=1}^{\infty} F_j\right) \\
&= \mu\left(\left(\bigcap_{j=1}^{\infty} F_j^C\right)^C\right) \\
&= \mu\left(X \setminus \left(\bigcap_{j=1}^{\infty} F_j^C\right)\right) \\
&= \mu(X) - \mu\left(\bigcap_{j=1}^{\infty} F_j^C\right) \\
&= \mu(X) - \lim_{j \rightarrow \infty} \mu(F_j^C) \\
&= \mu(X) - \lim_{j \rightarrow \infty} \mu(X \setminus F_j) \\
&= \mu(X) - \lim_{j \rightarrow \infty} (\mu(X) - \mu(F_j)) \\
&= \mu(X) - \lim_{j \rightarrow \infty} \mu(X) + \lim_{j \rightarrow \infty} \mu(F_j) \\
&= \lim_{j \rightarrow \infty} \mu(F_j) \\
&= \sum_{j=1}^{\infty} \mu(E_j).
\end{aligned}$$

□

Exercise 12. Let $(X, \mathcal{M}, \mu)$ be a finite measure space.

- a. If $E, F \in \mathcal{M}$ and $\mu(E \triangle F) = 0$, then $\mu(E) = \mu(F)$.
- b. Say that $E \sim F$ if $\mu(E \triangle F) = 0$; then $\sim$ is an equivalence relation on $\mathcal{M}$.
- c. For $E, F \in \mathcal{M}$, define $\rho(E, F) = \mu(E \triangle F)$. Then $\rho(E, G) \leq \rho(E, F) + \rho(F, G)$, and hence ρ defines a metric on the space $\mathcal{M}/\sim$ of equivalence classes.

Proof.

- a. Observe that

$$0 = \mu(E \triangle F) = \mu(E \cup F) - \mu(E \cap F),$$

so

$$\begin{aligned}
\mu(E \cap F) &= \mu(E \cup F) \\
&= \mu((E \setminus F) \cup F) \\
&= \mu(E \setminus F) + \mu(F) \\
&= \mu(F)
\end{aligned}$$

since $E \setminus F \subset E \triangle F$, and thus $\mu(E \setminus F) = 0$ by monotonicity. Swapping E and F in the above calculation reveals that $\mu(E \cap F) = \mu(E)$ as well, so $\mu(F) = \mu(E)$.

- b.** Surely given $E \in \mathcal{M}$, $\mu(E \triangle E) = \mu(\emptyset) = 0$, so $E \sim E$. Given also $F \in \mathcal{M}$ such that $E \sim F$, then since $E \triangle F = F \triangle E$, $\mu(F \triangle E) = 0$ so that $F \sim E$. Finally, if we have $G \in \mathcal{M}$ such that $F \sim G$, then notice that

$$E \triangle G \subset (E \triangle F) \cup (F \triangle G).$$

The easiest way to see this is with a venn diagram (shoot me), but it can also be shown with some slightly painful set theory. From this, we use monotonicity and subadditivity (respectively) to get

$$\begin{aligned}
\mu(E \triangle G) &\leq \mu((E \triangle F) \cup (F \triangle G)) \\
&\leq \mu(E \triangle F) + \mu(F \triangle G) \\
&= 0,
\end{aligned}$$

so $\mu(E \triangle G) = 0$.

- c.** By the definition of $\sim$, $\rho(E, F) = 0$ if and only if $E \sim F$. Also, $\rho(E, F) = \mu(E \triangle F) = \mu(F \triangle E) = \rho(F, E)$, so the problem statement is justified in that we need only prove the triangle inequality.

The following is similar to the end of **b**, but is reproduced here without the specific choices of E, F, G used in **b** and adapted for the problem at hand. Given *any* $E, F, G \in \mathcal{M}$, we have

$$\begin{aligned}
\rho(E, G) &= \mu(E \triangle G) \\
&\leq \mu((E \triangle F) \cup (F \triangle G)) \\
&\leq \mu(E \triangle F) + \mu(F \triangle G) \\
&= \rho(E, F) + \rho(F, G)
\end{aligned}$$

with the second line following from monotonicity and the third from subadditivity.

□

Exercise 13. Every σ -finite measure is semifinite.

Proof. Given a σ -finite measure space $(X, \mathcal{M}, \mu)$, if μ is finite, then semifiniteness is obvious. Assume, then, that $\mu(X) = \infty$, and let $E \in \mathcal{M}$ such that $\mu(E) = \infty$. Also, choose $\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{M}$ such that $\mu(E_j) < \infty$ for each j and $\bigcup_1^\infty E_j = X$. Then $\bigcup_1^\infty (E \cap E_j) = E$ and by monotonicity $\mu(E \cap E_j) \leq \mu(E_j) < \infty$ for all j . It cannot be the case that each $E \cap E_j$ has measure 0 since

$$\infty = \mu\left(\bigcup_1^\infty (E \cap E_j)\right) \leq \sum_1^\infty \mu(E \cap E_j),$$

so we can choose $k \in \mathbb{N}$ such that $0 < \mu(E \cap E_k) < \infty$ as desired. $\square$

Exercise 14. If μ is a semifinite measure and $\mu(E) = \infty$, for any $C > 0$ there exists $F \subset E$ with $C < \mu(F) < \infty$.

[NOTE FROM CONNOR: I'm pretty proud of this proof.]

Proof. Let $(X, \mathcal{M}, \mu)$ be the measure space in question. Given $E \in \mathcal{M}$ such that $\mu(E) = \infty$, define $A = \{F : F \in \mathcal{M}, F \subset E, 0 < \mu(F) < \infty\}$ and $B = \{\mu(F) : F \in A\}$. Now assume (contradicting the hypothesis) that B is (finitely) bounded from above. Then B has a least upper bound b .

Choose $F_1, F_2, \dots \in A$ such that $\{\mu(F_j)\}_{j \in \mathbb{N}}$ converges to b . This is possible because b is a limit point of B . Then define $G_j = \bigcup_{k=1}^j F_k$ for all $j \in \mathbb{N}$. $\{G_j\}$ is an ascending chain of finitely measured sets, and $G_j \subset E$ for all j , so $\bigcup_{j=1}^\infty G_j \subset E$ and $\mu\left(\bigcup_{j=1}^\infty G_j\right) = \lim_{j \rightarrow \infty} \mu(G_j)$. The sequence $\{\mu(G_j)\}$ is monotonically increasing due to monotonicity of μ , and it is bounded from above because $\mu(G_j) \leq b$ for any j , so it does converge. In fact, $\mu(F_j) \leq \mu(G_j) \leq b$ for any j , which means that $\{\mu(G_j)\}$ is sandwiched between $\{\mu(F_j)\}$ (which converges to b) and the constant sequence $\{b\}$ (which also converges to b), so $\{\mu(G_j)\}$ must also converge to b . Thus $\mu\left(\bigcup_{j=1}^\infty G_j\right) = b$.

Given this, we see that $\mu\left(E \setminus \bigcup_{j=1}^\infty G_j\right) = \infty$, so there is a set $G \subset E \setminus \bigcup_{j=1}^\infty G_j$ such that $0 < \mu(G) < \infty$. Then $H = G \cup \bigcup_{j=1}^\infty G_j$ is such that $H \subset E$, $0 < \mu(H) < \infty$, and $\mu(H) > b$. Thus $H \in A$ and $\mu(H) \in B$, which contradicts our assumption of b being a supremum of B . $\square$

Exercise 15. Given a measure μ on $(X, \mathcal{M})$, define μ_0 on $\mathcal{M}$ by $\mu_0(E) = \sup\{\mu(F) : F \subset E \text{ and } \mu(F) < \infty\}$.

- a. μ_0 is a semifinite measure. It is called the *semifinite part* of μ .
- b. If μ is semifinite, then $\mu = \mu_0$. (Use Exercise 14.)
- c. There is a measure ν on $\mathcal{M}$ (in general, not unique) which assumes only the values 0 and ∞ such that $\mu = \mu_0 + \nu$.

Proof.

- a. Clearly $\mu_0(\emptyset) = 0$ since the only subset of $\emptyset$ is $\emptyset$, whose measure is 0. Given a disjoint collection $\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{M}$, we have

$$\begin{aligned} \mu_0\left(\bigcup_{j=1}^{\infty} E_j\right) &= \sup \left\{ \mu(F) : F \subset \bigcup_{j=1}^{\infty} E_j \text{ and } \mu(F) < \infty \right\} \\ &= \sup \left\{ \sum_{k=1}^{\infty} \mu(F \cap E_k) : F \subset \bigcup_{j=1}^{\infty} E_j \text{ and } \mu(F) < \infty \right\} \\ &= \sum_{k=1}^{\infty} \sup \left\{ \mu(F \cap E_k) : F \subset \bigcup_{j=1}^{\infty} E_j \text{ and } \mu(F) < \infty \right\} \end{aligned}$$

CHECK THIS SET EQUALITY CAREFULLY

$$\begin{aligned} &= \sum_{k=1}^{\infty} \sup \{ \mu(F) : F \subset E_k \text{ and } \mu(F) < \infty \} \\ &= \sum_{k=1}^{\infty} \mu_0(E_k). \end{aligned}$$

So μ_0 is a measure.

Given any $E \in \mathcal{M}$ such that $\mu_0(E) = \infty$, by definition we see that

$$\infty = \sup \{ \mu(F) : F \subset E \text{ and } \mu(F) < \infty \},$$

so certainly there exists measurable $F \subset E$ such that $0 < \mu(F) < \infty$. Thus μ_0 is semifinite.

- b. Note that if $\mu(E)$ is finite, then clearly $\mu_0(E) = \mu(E)$, so we need only check agreement on measurable sets where $\mu(E) = \infty$. Given, then, $E \in \mathcal{M}$ such that $\mu(E) = \infty$, by Exercise 14 we know that

$$\{ \mu(F) : F \subset E \text{ and } \mu(F) < \infty \}$$

is unbounded from above, and thus its supremum is ∞ . Therefore

$$\mu_0(E) = \infty = \mu(E).$$

- c. Define ν so that given $E \in \mathcal{M}$, $\nu(E) = 0$ if $\mu(E) < \infty$ and $\nu(E) = \infty$ if $\mu(E) = \infty$. As observed in **b**, $\mu(E) = \mu_0(E)$ when $\mu(E) < \infty$, so

$$\mu_0(E) + \nu(E) = \mu_0(E) = \mu(E)$$

in that case. In case $\mu(E) = \infty$, then no matter the value of $\mu_0(E)$,

$$\mu_0(E) + \nu(E) = \infty = \mu(E).$$

□

Exercise 16. Let $(X, \mathcal{M}, \mu)$ be a measure space. A set $E \subset X$ is called **locally measurable** if $E \cap A \in \mathcal{M}$ for all $A \in \mathcal{M}$ such that $\mu(A) < \infty$. Let $\tilde{\mathcal{M}}$ be the collection of all locally measurable sets. Clearly $\mathcal{M} \subset \tilde{\mathcal{M}}$; then if $\mathcal{M} = \tilde{\mathcal{M}}$, then μ is called **saturated**.

- a. If μ is σ -finite, then μ is saturated.
- b. $\tilde{\mathcal{M}}$ is a σ -algebra.
- c. Define $\tilde{\mu}$ on $\tilde{\mathcal{M}}$ by $\tilde{\mu}(E) = \mu(E)$ if $E \in \mathcal{M}$ and $\tilde{\mu}(E) = \infty$ otherwise. Then $\tilde{\mu}$ is a saturated measure on $\tilde{\mathcal{M}}$, called the **saturation** of μ .
- d. If μ is complete, so is $\tilde{\mu}$.
- e. Suppose that μ is semifinite. For $E \in \tilde{\mathcal{M}}$, define $\underline{\mu}(E) = \sup \{\mu(A) : A \in \mathcal{M} \text{ and } A \subset E\}$. Then $\underline{\mu}$ is a saturated measure on $\tilde{\mathcal{M}}$ that extends μ .
- f. Let X_1, X_2 be disjoint uncountable sets, $X = X_1 \cup X_2$, and $\mathcal{M}$ the σ -algebra of countable or co-countable sets in X . Let μ_0 be counting measure on $\mathcal{P}(X_1)$, and define μ on $\mathcal{M}$ by $\mu(E) = \mu_0(E \cap X_1)$. Then μ is a measure on $\mathcal{M}$, $\tilde{\mathcal{M}} = \mathcal{P}(X)$, and in the notation of parts (c) and (e), $\tilde{\mu} \neq \underline{\mu}$.

Proof.

- a. Let $E \subset X$ be locally measurable and let $\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{M}$ such that $\mu(E_j) < \infty$ for each j and $\bigcup_1^\infty E_j = X$. Then $E = \bigcup_1^\infty (E_j \cap E)$, and because E is locally measurable, each $E_j \cap E$ is measurable, so E is a countable union of measurable sets. Thus E is measurable.
- b. Given $\{E_j\}_{j \in \mathbb{N}} \subset \tilde{\mathcal{M}}$ and $A \in \mathcal{M}$ such that $\mu(A) < \infty$,

$$\left(\bigcup_1^\infty E_j \right) \cap A = \bigcup_1^\infty (E_j \cap A),$$

so $(\bigcup_1^\infty E_j) \cap A$ is the countable union of measurable sets and is thus measurable itself.

Also, given $E \in \tilde{\mathcal{M}}$ and A as before,

$$E^C \cap A = A \setminus (E \cap A).$$

Since $E \cap A \in \mathcal{M}$ and $\mathcal{M}$ is closed under differences, $E^C \cap A \in \mathcal{M}$.

Thus $\tilde{\mathcal{M}}$ is a σ -algebra.

- c. First we show that $\tilde{\mu}$ is a measure. Since $\emptyset \in \mathcal{M}$, $\tilde{\mu}(\emptyset) = \mu(\emptyset) = 0$. Also, let $\{E_j\}_{j \in \mathbb{N}} \subset \tilde{\mathcal{M}}$ be a disjoint collection. If $\bigcup_1^\infty E_j \in \mathcal{M}$, then since each

E_j is locally measurable, $E_k = E_k \cap \bigcup_1^\infty E_j \in \mathcal{M}$ for each k . Then

$$\begin{aligned}\tilde{\mu}\left(\bigcup_1^\infty E_j\right) &= \mu\left(\bigcup_1^\infty E_j\right) \\ &= \sum_1^\infty \mu(E_j) \\ &= \sum_1^\infty \tilde{\mu}(E_j).\end{aligned}$$

On the other hand, if $\bigcup_1^\infty E_j \notin \mathcal{M}$, then for some $k \in \mathbb{N}$, $E_k \notin \mathcal{M}$. Thus

$$\begin{aligned}\tilde{\mu}\left(\bigcup_1^\infty E_j\right) &= \infty \\ &= \sum_1^\infty \tilde{\mu}(E_j)\end{aligned}$$

since $\tilde{\mu}(E_k) = \infty$.

Now we show that $\tilde{\mu}$ is saturated. Let $E \in X$ locally measurable over $\tilde{\mu}$. We want to show that $E \in \tilde{\mathcal{M}}$. Let $A \in \mathcal{M}$ such that $\mu(A) < \infty$. Then since $A \in \tilde{\mathcal{M}}$, $E \cap A \in \tilde{\mathcal{M}}$, and by monotonicity

$$\tilde{\mu}(E \cap A) < \tilde{\mu}(A) = \mu(A) < \infty.$$

Thus since $\tilde{\mu}(E \cap A)$ is finite, $E \cap A \in \mathcal{M}$. Therefore $E \in \tilde{\mathcal{M}}$.

- d. Let $F \subset N$ for some $N \in \tilde{\mathcal{M}}$ where $\tilde{\mu}(N) = 0$. Then $N \in \mathcal{M}$ since $\tilde{\mu}(N)$ is finite. Since μ is complete, $F \in \mathcal{M} \subset \tilde{\mathcal{M}}$. Thus $\tilde{\mu}$ is complete.
- e. First we prove that μ is a measure. Since $\emptyset$ is the only subset of $\emptyset$ and $\mu(\emptyset) = 0$, $\underline{\mu}(\emptyset) = \sup\{0\} = 0$. Also, given a disjoint collection $\{E_j\}_{j \in \mathbb{N}} \subset$

$\tilde{\mathcal{M}}$,

$$\begin{aligned}
\underline{\mu}\left(\bigcup_1^\infty E_j\right) &= \sup\left\{\mu(A) : A \in \mathcal{M} \text{ and } A \subset \bigcup_1^\infty E_j\right\} \\
&= \sup\left\{\mu\left(\bigcup_{k=1}^\infty A \cap E_k\right) : A \in \mathcal{M} \text{ and } A \subset \bigcup_1^\infty E_j\right\} \\
&= \sup\left\{\sum_{k=1}^\infty \mu(A \cap E_k) : A \in \mathcal{M} \text{ and } A \subset \bigcup_1^\infty E_j\right\} \\
&= \sum_{k=1}^\infty \sup\left\{\mu(A \cap E_k) : A \in \mathcal{M} \text{ and } A \subset \bigcup_1^\infty E_j\right\} \\
&= \sum_{k=1}^\infty \sup\{\mu(A) : A \in \mathcal{M} \text{ and } A \subset E_k\} \\
&= \sum_{k=1}^\infty \underline{\mu}(E_k).
\end{aligned}$$

Thus $\underline{\mu}$ is a measure.

To show that it extends μ is easy since by monotonicity, any subset of some $E \in \mathcal{M}$ has measure less than or equal to that of E , so

$$\underline{\mu}(E) = \sup\{\mu(A) : A \in \mathcal{M} \text{ and } A \subset E\} = \mu(E).$$

Now we just need to show saturation. In particular, if $E \in X$ is locally measurable over $\underline{\mu}$, then we need to show that $E \in \tilde{\mathcal{M}}$. Thus we take $A \in \mathcal{M}$ such that $\mu(A) < \infty$ and see that since $A \in \tilde{\mathcal{M}}$, $E \cap A \in \tilde{\mathcal{M}}$. Then $E \cap A$ is locally measurable over μ , so $E \cap A = (E \cap A) \cap A \in \mathcal{M}$. Therefore $\underline{\mu}$ is saturated.

- f. First we show that μ is a measure on $\mathcal{M}$. We see that $\mu(\emptyset) = \mu_0(\emptyset \cap X_1) = \mu_0(\emptyset) = 0$. Let $\{E_j\}_{j \in \mathbb{N}} \subset \mathcal{M}$ be disjoint. Then

$$\begin{aligned}
\mu\left(\bigcup_1^\infty E_j\right) &= \mu_0\left(\left(\bigcup_1^\infty E_j\right) \cap X_1\right) \\
&= \mu_0\left(\bigcup_1^\infty (E_j \cap X_1)\right) \\
&= \sum_1^\infty \mu_0(E_j \cap X_1) \\
&= \sum_1^\infty \mu(E_j).
\end{aligned}$$

Next we want to show that $\tilde{\mathcal{M}} = \mathcal{P}(X)$, or in other words, that every subset of X is locally measurable. Let $E \subset X$ and $A \in \mathcal{M}$ such that $\mu(A) < \infty$. Notice that $A \cap X_1$ must be finite since $\mu_0(A \cap X_1) < \infty$. Thus A^C is uncountable because X_1 is uncountable and $A^C \cap X_1 = X_1 \setminus (A \cap X_1)$. This means A cannot be co-countable, so since $A \in \mathcal{M}$, it must be countable. Seeing this, it is obvious that $E \cap A$ is countable because $E \cap A \subset A$, so $E \cap A \in \mathcal{M}$.

Finally, to see that $\tilde{\mu} \neq \underline{\mu}$, observe that $X_2 \notin \mathcal{M}$ because X_2 is uncountable and $X_2^C = X_1$ is uncountable. Then $\tilde{\mu}(X_2) = \infty$ because $X_2^2 \notin \mathcal{M}$, but $\mu(X_2) = \mu_0(X_2 \cap X_1) = \mu_0(\emptyset) = 0$, so any subset of X_2 also has measure 0. Thus

$$\underline{\mu}(X_2) = \sup \{ \mu(A) : A \in \mathcal{M} \text{ and } A \subset X_2 \} = \sup \{ 0 \} = 0.$$

We conclude that since $\tilde{\mu}(X_2) = \infty \neq 0 = \underline{\mu}(X_2)$, $\tilde{\mu} \neq \underline{\mu}$.

□

Exercise 17. If μ^* is an outer measure on X and $\{A_j\}_1^\infty$ is a sequence of disjoint μ^* -measurable sets, then $\mu^*(E \cap (\bigcup_1^\infty A_j)) = \sum_1^\infty \mu^*(E \cap A_j)$ for any $E \subset X$.

Proof. Notice that

$$\mu^* \left(E \cap \left(\bigcup_1^\infty A_j \right) \right) = \mu^* \left(\bigcup_1^\infty E \cap A_j \right) \leq \sum_1^\infty \mu^*(E \cap A_j),$$

so to prove equality, we need only show that $\mu^*(E \cap \bigcup_1^\infty A_j) \geq \sum_1^\infty \mu^*(E \cap A_j)$.

Given $n \in \mathbb{N}$, define $E_n = E \cap A_n$ and $B_n = \bigcup_{j=1}^n A_j$. Then

$$\begin{aligned} \mu^*(E \cap B_n) &= \mu^*(E \cap B_n \cap A_n) + \mu^*(E \cap B_n \cap A_n^C) \\ &= \mu^*(E \cap A_n) + \mu^*(E \cap B_{n-1}) \end{aligned}$$

when $n > 1$. If $n = 1$, then clearly $\mu^*(E \cap B_1) = \mu^*(E \cap A_1)$ by definition. By induction, we get $\mu^*(E \cap B_n) = \sum_{j=1}^n \mu^*(E \cap A_j)$. Thus

$$\lim_{n \rightarrow \infty} \mu^*(E \cap B_n) = \sum_{j=1}^\infty \mu^*(E \cap A_j). \quad (1)$$

Since each $E \cap B_n \subset E \cap \bigcup_{j=1}^\infty A_j$, each $\mu^*(E \cap B_n) \leq \mu^*(E \cap \bigcup_{j=1}^\infty A_j)$, so we see that

$$\mu^*(E \cap \bigcup_{j=1}^\infty A_j) \geq \lim_{n \rightarrow \infty} \mu^*(E \cap B_n).$$

Combining with (1) yields $\mu^*(E \cap \bigcup_{j=1}^\infty A_j) \geq \sum_{j=1}^\infty \mu^*(E \cap A_j)$. □

Exercise 18. Let $\mathcal{A} \subset \mathcal{P}(X)$ be an algebra, $\mathcal{A}_\sigma$ the collection of countable unions of sets in $\mathcal{A}$, and $\mathcal{A}_{\sigma\delta}$ the collection of countable intersections of sets in $\mathcal{A}_\sigma$. Let μ_0 be a premeasure on $\mathcal{A}$ and μ^* the induced outer measure.

- a. For any $E \subset X$ and $\epsilon > 0$ there exists $A \in \mathcal{A}_\sigma$ with $E \subset A$ and $\mu^*(A) \leq \mu^*(E) + \epsilon$.
- b. If $\mu^*(E) < \infty$, then E is μ^* -measurable iff there exists $B \in \mathcal{A}_{\sigma\delta}$ with $E \subset B$ and $\mu^*(B \setminus E) = 0$.
- c. If μ_0 is σ -finite, the restriction $\mu^*(E) < \infty$ in (b) is superfluous.

Proof.

- a. By the definition of μ^* , we know there exists $\{A_j\}_1^\infty \subset \mathcal{A}$ such that $E \subset \bigcup_1^\infty A_j = A \in \mathcal{A}_\sigma$ where $\sum_1^\infty \mu_0(A_j) < \mu^*(E) + \epsilon$. Clearly $A \subset \bigcup_1^\infty A_j$, so by following the definition of $\mu^*(A)$, we get $\mu^*(A) \leq \sum_1^\infty \mu_0(A_j)$. Combining these two inequalities, we get

$$\mu^*(A) \leq \sum_1^\infty \mu_0(A_j) < \mu^*(E) + \epsilon$$

as desired.

- b. If E is μ^* -measurable, we define B as follows. For each $j \in \mathbb{N}$, let $B_j = \bigcup_{k=1}^\infty B_{jk}$ such that each $B_{jk} \in \mathcal{A}$, $E \subset B_j$, and $\sum_{k=1}^\infty \mu_0(B_{jk}) < \mu^*(E) + \frac{1}{j}$. Let $B = \bigcap_{j=1}^\infty B_j$. Then by definition $B \in \mathcal{A}_{\sigma\delta}$, and since $E \subset B_j$ for each j , $E \subset B$. Thus $\mu^*(E) \leq \mu^*(B)$, and fixing j we get $B \subset B_j$, $\mu^*(E) \leq \mu^*(B) \leq \mu^*(E) + \frac{1}{j}$. Since this is true for every j , we have $\mu^*(E) = \mu^*(B)$. Because E is μ^* -measurable,

$$\begin{aligned} \mu^*(B) &= \mu^*(B \cap E) + \mu^*(B \cap E^C) \\ &= \mu^*(E) + \mu^*(B \setminus E). \end{aligned}$$

Then

$$\mu^*(E) = \mu^*(B) = \mu^*(E) + \mu^*(B \setminus E),$$

so $\mu^*(B \setminus E) = 0$ as desired since $\mu^*(E) < \infty$.

Conversely, assume there exists $B \in \mathcal{A}_{\sigma\delta}$ such that $E \subset B$ and $\mu^*(B \setminus E) = 0$. Since B is a countable intersection of countable unions of elements of $\mathcal{A}$ which are each μ^* -measurable (Proposition 1.3), B is μ^* -measurable. Given $F \subset X$, notice that

$$\begin{aligned} \mu^*(F \cap B^C) &= \mu^*(F \cap (B \cup E)^C) \\ &= \mu^*(F \cap B^C \cap E^C) \\ &= \mu^*(F \cap E^C \cap B^C) \\ &= \mu^*(F \cap E^C) - \mu^*(F \cap E^C \cap B) \\ &= \mu^*(F \cap E^C) - \mu^*(F \cap (B \setminus E)) \\ &= \mu^*(F \cap E^C). \end{aligned}$$

Then we get

$$\begin{aligned}\mu^*(F) &= \mu^*(F \cap B) + \mu^*(F \cap B^C) \\ &\geq \mu^*(F \cap E) + \mu^*(F \cap E^C),\end{aligned}$$

which shows that E is μ^* -measurable.

- c. We need to prove the forward direction with the hypothesis that μ_0 is σ -finite instead of $\mu^*(E) < \infty$. Consider first that $X = \bigcup_1^\infty A_j$, a union which we can assume to be disjoint, where each $A_j \in \mathcal{A}$ such that $\mu_0(A_j) < \infty$. Note that each A_j is also μ^* -measurable since it is in $\mathcal{A}$.

For any $j, n \in \mathbb{N}$, define $E_j = E \cap A_j$ and choose $E_{jn} \in \mathcal{A}_\sigma$ such that $E_j \subset E_{jn} \subset A_j$ and $\mu^*(E_{jn}) \leq \mu^*(E_j) + \frac{1}{2^j n}$, using part a. Note that since E_j is μ^* -measurable,

$$\begin{aligned}\mu(E_{jn}) &= \mu^*(E_{jn} \cap E_j) + \mu^*(E_{jn} \cap E_j^C) \\ &= \mu^*(E_j) + \mu^*(E_{jn} \setminus E_j),\end{aligned}$$

so $\mu(E_{jn}) - \mu^*(E_j) = \mu^*(E_{jn} \setminus E_j)$. Also, by the above inequality, $\mu^*(E_{jn} \setminus E_j) \leq \frac{1}{2^j n}$.

Let $B_n = \bigcup_{j=1}^\infty E_{jn}$ and $B = \bigcap_1^\infty B_n$. Since each $B_n \in \mathcal{A}_\sigma$, $B \in \mathcal{A}_{\sigma\delta}$. Also, since $E \subset B_n$ for each n , $E \subset B$. Finally, we compute $\mu^*(B \setminus E)$:

$$\begin{aligned}\mu^*(B_n \setminus E) &= \mu^*(B_n \cap E^C) \\ &= \sum_{j=1}^\infty \mu^*(E_{jn} \cap E^C) \\ &= \sum_{j=1}^\infty \mu^*(E_{jn} \cap E_j^C) \\ &= \sum_{j=1}^\infty \mu^*(E_{jn} \setminus E_j) \\ &\leq \sum_{j=1}^\infty \frac{1}{2^j n} \\ &= \frac{1}{n}\end{aligned}$$

for all n . Thus since $(B \setminus E) \subset (B_n \setminus E)$,

$$\mu^*(B \setminus E) \leq \mu^*(B_n \setminus E) \leq \frac{1}{n}.$$

This is true for all n , so $\mu^*(B \setminus E) = 0$.

□

Exercise 19. Let μ^* be an outer measure on X induced from a finite premeasure μ_0 . If $E \subset X$, define the **inner measure** of E to be $\mu_*(E) = \mu_0(X) - \mu^*(E^C)$. Then E is μ^* -measurable iff $\mu^*(E) = \mu_*(E)$. (Use Exercise 18.)

Proof. For the forward direction, if $E \subset X$ is μ^* -measurable, then

$$\begin{aligned}\mu_*(E) &= \mu_0(X) - \mu^*(E^C) \\ &= \mu^*(X) - \mu^*(E^C) \\ &= \mu^*(X \cap E) + \mu^*(X \cap E^C) - \mu^*(E^C) \\ &= \mu^*(E) + \mu^*(E^C) - \mu^*(E^C) \\ &= \mu^*(E).\end{aligned}$$

For the other direction, we start with

$$\begin{aligned}\mu^*(E) &= \mu_0(X) - \mu^*(E^C) \\ &= \mu^*(X) - \mu^*(E^C).\end{aligned}$$

Since $\mu^*(E) = \mu^*(X \cap E) = \mu^*(X \setminus E^C)$, we apply the above and get

$$\mu^*(X \setminus E^C) = \mu^*(X) - \mu^*(E^C).$$

We proceed to find a descending chain of sets in $\mathcal{A}_\sigma$ whose outer measures approach that of E . By 18a, we can find for each $n \in \mathbb{N}$ a $B_n \in \mathcal{A}_\sigma$ such that $E \subset B_n$ and $\mu^*(B_n) \leq \mu^*(E) + \frac{1}{n}$. Furthermore, we choose each B_n such that $B_{n+1} \subset B_n$. The existence of such a sequence is evidenced by the fact that if $B_{n+1} \supset B_n$, then $B_{n+1} \cap B_n \subset B_n$, $B_{n+1} \cap B_n \supset E$, and by monotonicity, $\mu^*(B_{n+1} \cap B_n) \leq \mu^*(B_{n+1}) \leq \mu^*(E) + \frac{1}{n}$.

We then define $B = \bigcap_{n=1}^{\infty} B_n$, which is clearly in $\mathcal{A}_{\sigma\delta}$ and contains E . It follows immediately that since $\{B_n\}_1^{\infty}$ is a descending chain of μ^* -measurable sets and $\mu^*|_{\mathcal{M}}$ is a measure where $\mathcal{M} = \{E \subset X \mid E \text{ is } \mu^*\text{-measurable}\}$, then $\mu^*(B) = \lim_{n \rightarrow \infty} \mu^*(B_n) = \mu^*(E)$. To satisfy the condition from 18b, we need only prove that $\mu^*(B \setminus E) = 0$.

For each n , since B_n is μ^* -measurable, we have

$$\begin{aligned}\mu^*(E^C) &= \mu^*(E^C \cap B_n) + \mu^*(E^C \cap B_n^C) \\ &= \mu^*(B_n \setminus E) + \mu^*((E \cup B_n)^C) \\ &= \mu^*(B_n \setminus E) + \mu^*(B_n^C),\end{aligned}$$

so

$$\begin{aligned}\mu^*(B_n \setminus E) &= \mu^*(E^C) - \mu^*(B_n^C) \\ &= \mu^*(X) - \mu^*(E) - \mu^*(X) + \mu^*(B_n) \\ &= \mu^*(B_n) - \mu^*(E)\end{aligned}$$

where the second equality applies the forward direction of the present theorem to B_n^C . We can now conclude that

$$\mu^*(B_n \setminus E) = \mu^*(B_n) - \mu^*(E) \leq \mu^*(E) + \frac{1}{n} - \mu^*(E) = \frac{1}{n}.$$

Since $(B \setminus E) \subset (B_n \setminus E)$, $\mu^*(B \setminus E) \leq \mu^*(B_n \setminus E) \leq \frac{1}{n}$. This is true for all n , so $\mu^*(B \setminus E) = 0$, and we can use 18b to conclude that E is μ^* -measurable. $\square$

Exercise 20. Let μ^* be an outer measure on X , $\mathcal{M}^*$ the σ -algebra of μ^* -measurable-sets, $\bar{\mu} = \mu^*|_{\mathcal{M}^*}$, and μ^+ the outer measure induced by $\bar{\mu}$ as in (1.12) (with $\bar{\mu}$ and $\mathcal{M}^*$ replacing μ_0 and $\mathcal{A}$).

- a. If $E \subset X$, we have $\mu^*(E) \leq \mu^+(E)$, with equality iff there exists $A \in \mathcal{M}^*$ with $A \supset E$ and $\mu^*(A) = \mu^*(E)$.
- b. If μ^* is induced from a premeasure, then $\mu^* = \mu^+$. (Use Exercise 18a.)
- c. If $X = \{0, 1\}$, there exists an outer measure μ^* on X such that $\mu^* \neq \mu^+$.

Proof.

- a. If $E \subset X$, define

$$C_E = \left\{ \sum_1^\infty \mu^*(M_j) : M_j \in \mathcal{M}^* \text{ and } E \subset \bigcup_1^\infty M_j \right\}$$

so that $\mu^+(E) = \inf C_E$ by definition. Then given $\{M_j\}_1^\infty \subset \mathcal{M}^*$ such that $E \subset \bigcup_1^\infty M_j$, then

$$\mu^*(E) \leq \mu^* \left(\bigcup_1^\infty M_j \right) \leq \sum_1^\infty \mu^*(M_j).$$

Thus $\mu^*(E)$ is a lower bound of C_E , so $\mu^*(E) \leq \inf(C_E) = \mu^+(E)$.

Now if there exists $A \in \mathcal{M}^*$ such that $A \supset E$ and $\mu^*(A) = \mu^*(E)$, then the collection $\{A_j\}_1^\infty$ where $A_1 = A$ and $A_n = \emptyset$ for $n > 1$ is such that $E \subset \bigcup_1^\infty A_j$ and

$$\sum_1^\infty \mu^*(A_j) = \mu^*(A) = \mu^*(E).$$

Thus $\mu^*(E) \in C_E$ and $\mu^*(E)$ is a lower bound of C_E , so $\mu^*(E) = \inf C_E = \mu^+(E)$.

On the other hand, if $\mu^*(E) = \mu^+(E)$, then choose for $n \in \mathbb{N}$ an $A_n \in \mathcal{M}^*$ such that $E \subset A_n$ and $\mu^*(A_n) \leq \mu^+(E) + \frac{1}{n} = \mu^*(E) + \frac{1}{n}$ (Exercise 18a). Then clearly $\mu^*(\bigcap_1^\infty A_n) = \mu^*(E)$, and $\bigcap_1^\infty A_n \in \mathcal{M}^*$ since $\mathcal{M}^*$ is a σ -algebra.

- b. Let $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ be the premeasure which induces μ^* . Given $E \subset X$, by Exercise 18a we can choose for any $n \in \mathbb{N}$ an $A_n \in \mathcal{A}_\sigma$ such that $E \subset A_n$ and $\mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}$. Let $A = \bigcap_1^\infty A_n$. Then $\mu^*(A) \leq \mu^*(A_n) \leq \mu^*(E) + \frac{1}{n}$ for all n , so $\mu^*(A) = \mu^*(E)$. Since any element of $\mathcal{A}$ is μ^* -measurable and A is a countable intersection of countable unions of elements of $\mathcal{A}$, $A \in \mathcal{M}^*$. Thus by part a of the present exercise, $\mu^*(E) = \mu^+(E)$, and since E varies over $\mathcal{P}(X)$, $\mu^* = \mu^+$.
- c. Consider the outer measure μ^* defined by

$$\begin{aligned}\emptyset &\mapsto 0 \\ \{0\} &\mapsto 2 \\ \{1\} &\mapsto 2 \\ X &\mapsto 3.\end{aligned}$$

The proof that this is an outer measure is routine. Notice that $\{0\}$ is not μ^* -measurable since

$$\mu^*(X \cap \{0\}) + \mu^*(X \cap \{1\}) = 4 > 3 = \mu^*(X).$$

Thus $\{1\}$ is also not μ^* -measurable, so $\mathcal{M}^* = \{X, \emptyset\}$. Since X is the only member of $\mathcal{M}$ containing $\{0\}$ or $\{1\}$, we can see that μ^+ is described by

$$\begin{aligned}\emptyset &\mapsto 0 \\ \{0\} &\mapsto 3 \\ \{1\} &\mapsto 3 \\ X &\mapsto 3.\end{aligned}$$

Thus $\mu^* \neq \mu^+$.

□

Exercise 21. Let μ^* be an outer measure induced from a premeasure and $\bar{\mu}$ the restriction of μ^* to the μ^* -measurable sets. Then $\bar{\mu}$ is saturated. (Use Exercise 18.)

Proof. Let $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ be the premeasure from which μ^* is induced. Given $E \subset X$ locally measurable (note that E^C is also locally measurable by 16b), we want to show that E is μ^* -measurable. We thus consider $F \subset X$ such that $\mu^*(F) < \infty$ and wish to show that $\mu^*(F) \geq \mu^*(F \cap E) + \mu^*(F \cap E^C)$.

By 18a, given $n \in \mathbb{N}$, we can find $B_n \in \mathcal{A}_\sigma$ such that $F \subset B_n$ and $\mu^*(B_n) \leq \mu^*(F) + \frac{1}{n}$. Fix $n \in \mathbb{N}$. Then B_n is μ^* -measurable, so we can use the finite additivity of $\bar{\mu}$ to get

$$\begin{aligned}\mu^*(B_n) &= \bar{\mu}(B_n) \\ &= \bar{\mu}(B_n \cap E) + \bar{\mu}(B_n \cap E^C) \\ &= \mu^*(B_n \cap E) + \mu^*(B_n \cap E^C) \\ &\leq \mu^*(F) + \frac{1}{n}.\end{aligned}$$

Since $(F \cap E) \subset (B_n \cap E)$ and $(F \cap E^C) \subset (B_n \cap E^C)$, monotonicity together with the final inequality from above shows us that

$$\begin{aligned}\mu^*(F \cap E) + \mu^*(F \cap E^C) &\leq \mu^*(B_n \cap E) + \mu^*(B_n \cap E^C) \\ &\leq \mu^*(F) + \frac{1}{n}.\end{aligned}$$

Letting n vary over $\mathbb{N}$, we conclude that

$$\mu^*(F \cap E) + \mu^*(F \cap E^C) \leq \mu^*(F).$$

Thus E is μ^* -measurable, so $\bar{\mu}$ is saturated. $\square$

Exercise 22. Let $(X, \mathcal{M}, \mu)$ be a measure space, μ^* the outer measure induced by μ according to (1.12), $\mathcal{M}^*$ the σ -algebra of μ^* -measurable sets, and $\bar{\mu} = \mu^*|_{\mathcal{M}^*}$.

- a. If μ is σ -finite, then $\bar{\mu}$ is the completion of μ . (Use Exercise 18.)
- b. In general, $\bar{\mu}$ is the saturation of the completion of μ . (See Exercises 16 and 21.)

Proof.

- a. Let $\mathcal{N} = \{N \in \mathcal{M} : \mu(N) = 0\}$ and

$$\overline{\mathcal{M}} = \{E \cup F : E \in \mathcal{M}, F \subset X \text{ such that } F \subset N \text{ for some } N \in \mathcal{N}\}.$$

We need to show that $\bar{\mu}$ is the unique complete measure extending μ on $\overline{\mathcal{M}}$. Thus we need to prove two things, the first of which is that $\mathcal{M}^* = \overline{\mathcal{M}}$, and the second of which is that $\bar{\mu}$ extends μ .

First, assume $M \in \mathcal{M}^*$, so M is μ^* -measurable. Then by 18c, we can find $B \in \mathcal{M}$ ($\mathcal{M} = \mathcal{M}_{\sigma\delta}$ since $\mathcal{M}$ is a σ -algebra) such that $M \subset B$ and $\mu^*(B \setminus M) = 0$. Since both B and M are μ^* -measurable, $B \setminus M$ is also μ^* -measurable, so we can repeat the process on $B \setminus M$ by finding $C \in \mathcal{M}$ such that $(B \setminus M) \subset C$ and $\mu^*(C \setminus (B \setminus M)) = 0$. Notice that $C \cap M$ is μ^* -measurable and

$$\begin{aligned}0 &= \mu^*(C \setminus (B \setminus M)) \\ &= \mu^*(C \cap (B \cap M^C)^C) \\ &= \mu^*(C \cap (B^C \cup M)) \\ &= \mu^*((C \cap B^C) \cup (C \cap M)),\end{aligned}$$

so by monotonicity, $\mu^*(C \cap M) = 0$. Thus we apply 18c one more time to get a $D \in \mathcal{M}$ such that $(C \cap M) \subset D$ and $\mu^*(D \setminus (C \cap M)) = 0$. Then

$$\mu^*(D) \leq \mu^*(D \setminus (C \cap M)) + \mu^*(C \cap M) = 0,$$

so in fact $D \in \mathcal{N}$, making $C \cap M$ a subset of an element of $\mathcal{N}$. Now all we need to know is that $(B \setminus C) \cup (C \cap M) = M$, and we will have shown that $M \in \overline{\mathcal{M}}$. The forward inclusion is easy, since if $x \in B \setminus C$ then $x \in M$ because C contains all the elements of B not in M , and clearly $C \cap M \subset M$. For the other direction, if $x \in M$, then either $x \in C$ and thus $x \in C \cap M$, or $x \notin C$, in which case since $x \in M \subset B$, $x \in B \setminus C$.

Now suppose $M \in \overline{\mathcal{M}}$, so $M = E \cup F$ for some $E \in \mathcal{M}$ and $F \in X$ such that $F \subset N$ for some $N \in \mathcal{N}$. Given $G \subset X$ such that $\mu^*(G) < \infty$, we want to show that

$$\mu^*(G) \geq \mu^*(G \cap M) + \mu^*(G \cap M^C).$$

This is straightforward:

$$\begin{aligned} \mu^*(G \cap M) + \mu^*(G \cap M^C) &= \mu^*(G \cap (E \cup F)) + \mu^*(G \cap (E \cup F)^C) \\ &= \mu^*((G \cap E) \cup (G \cap F)) + \mu^*(G \cap E^C \cap F^C) \\ &\leq \mu^*(G \cap E) + \mu^*(G \cap F) + \mu^*(G \cap E^C) \\ &= \mu^*(G \cap E) + \mu^*(G \cap E^C) \\ &= \mu^*(G) \end{aligned}$$

where the final equation follows because E is μ^* -measurable.

Finally, we need $\overline{\mu}(M) = \mu(M)$ for $M \in \mathcal{M}$. This follows immediately from (1.13), taking μ to be μ_0 and $\mathcal{M}$ to be $\mathcal{A}$.

- b.** Let $\overline{\mathcal{M}}$, $\mathcal{M}^*$, and $\mathcal{N}$ be defined as above, and $\widetilde{\mathcal{M}}$ the set of all locally measurable sets under $\overline{\mu}$ where $\overline{\mu}$ is the completion of μ . Then we need to show that $\mathcal{M}^* = \widetilde{\mathcal{M}}$ and that $\overline{\mu}(E) = \mu(E)$ if $E \in \overline{\mathcal{M}}$ and $\overline{\mu}(E) = \infty$ otherwise. Note that the proof of part **a** only used σ -finiteness to show that $\mathcal{M}^* \subset \overline{\mathcal{M}}$, so we can use the facts that $\overline{\mathcal{M}} \subset \mathcal{M}^*$ and $\mu^*(M) = \overline{\mu}(M)$ for all $M \in \mathcal{M}^*$.

First, let $M \in \mathcal{M}^*$. Then given $E \in \overline{\mathcal{M}}$, $\mu^*(E) = \mu^*(E \cap M) + \mu^*(E \cap M^C)$

Now let $M \in \widetilde{\mathcal{M}}$. Given $E \subset X$ such that $\mu^*(E) < \infty$ and $F \in \overline{\mathcal{M}}$,

$$\begin{aligned} \mu^*(E \cap M) + \mu^*(E \cap M^*) &= \dots \\ &= \dots \\ &\leq \dots \\ &= \dots \\ &= \mu^*(E) \end{aligned}$$

$$\begin{aligned} \mu^*(E) &= \mu^*(E \cap F \cap M) + \mu^*(E \cap (F \cap M)^C) \\ &= \mu^*(E \cap F \cap M) + \mu^*(E \cap (F^C \cup M^C)) \\ &= \mu^*(E \cap F \cap M) + \mu^*((E \cap F^C) \cup (E \cap M^C)) \\ &\leq \mu^*(E \cap F \cap M) + \mu^*(E \cap F^C) + \mu^*(E \cap M^C) \end{aligned}$$

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□

Exercise 25. Complete the proof of Theorem 1.19.

Proof. For each $j \in \mathbb{Z}$, let $E_j = E \cap (j, j+1]$. Then $\bigcup_1^\infty E_j = E$ and we can find open U_{jk} for each $j, k \in \mathbb{Z}$ such that $(\bigcap_{k=1}^\infty U_{jk}) \setminus N_j = E_j$ for some null set N_j . Then each $\bigcup_{j \in \mathbb{Z}} U_{jk}$ is open and contains E . Define $V = \bigcap_{k=1}^\infty \bigcup_{j \in \mathbb{Z}} U_{jk}$. Then V is a G_δ set containing E , and $V \setminus (V \setminus E) = E$, so we simply need to show that $\mu(V \setminus E) = 0$. Observe that

$$\begin{aligned} V \setminus E &= V \cap \left(\bigcup_{j \in \mathbb{Z}} E_j \right)^C \\ &= V \cap \left(\bigcap_{j \in \mathbb{Z}} E_j^C \right) \\ &= \left(\bigcap_{j \in \mathbb{Z}} V \cap E_j^C \right) \\ &= \left(\bigcap_{j \in \mathbb{Z}} V \setminus E_j \right) \\ &\subset V \setminus E_1 \\ &\subset N_1, \end{aligned}$$

so by monotonicity $\mu(V \setminus E) = 0$.

For $(a \implies c)$, define E_j as before and let H_{jk} be closed for each $j, k \in \mathbb{Z}$ such that each $E_j = (\bigcup_{k=1}^\infty H_{jk}) \cup N_j$ for some null N_j . Let $H = \bigcup_{j \in \mathbb{Z}} \bigcup_{k=1}^\infty H_{jk}$ and $N = \bigcup_{j \in \mathbb{Z}} N_j$. Then H is an F_σ set and $H \subset E$. Also, $\mu(N) = 0$ by subadditivity. Observe, then, that

$$\begin{aligned} E &= \bigcup_{j \in \mathbb{Z}} E_j \\ &= \bigcup_{j \in \mathbb{Z}} \bigcup_{k=1}^\infty H_{jk} \cup N_j \\ &= \left(\bigcup_{j \in \mathbb{Z}} \bigcup_{k=1}^\infty H_{jk} \right) \cup \bigcup_{j \in \mathbb{Z}} N_j \\ &= H \cup N. \end{aligned}$$

Then it becomes obvious that $E \setminus H \subset N$, so $\mu(E \setminus H) = 0$. □

Exercise 26. Prove Proposition 1.20. (Use Theorem 1.18.)

Proof. Given $\epsilon > 0$, choose $U \supset E$ be open such that $\mu(U) < \mu(E) + \epsilon/2$ and $H \subset E$ compact such that $\mu(H) > \mu(E) - \epsilon/2$. Then express U as a countable union of open intervals $\bigcup_1^\infty U_j = U$. Since $H \subset E \subset U$, $\{U_j\}_1^\infty$ is an open cover of H , so there is a finite subcover $\{U_j\}_1^n$ for some n after potential reordering. Define $A = \bigcup_1^n U_j$ and observe that $H \subset A \subset U$. Then

$$\mu(E) - \epsilon/2 < \mu(H) \leq \mu(A) \leq \mu(U) < \mu(E) + \epsilon/2.$$

We now observe that

$$E \cup A = (E \triangle A) \cup (E \cap A),$$

so

$$\mu(E \triangle A) = \mu(E \cup A) - \mu(E \cap A).$$

Clearly $H \subset (E \cup A) \subset U$ and $H \subset (E \cap A) \subset U$, so

$$\begin{aligned} \mu(E \triangle A) &= \mu(E \cup A) - \mu(E \cap A) \\ &< \mu(E) + \epsilon/2 - (\mu(E) - \epsilon/2) \\ &= \epsilon \end{aligned}$$

as desired. $\square$

Exercise 27. Prove Proposition 1.22a. (Show that if $x, y \in C$ and $x < y$, there exists $z \notin C$ such that $x < z < y$.)

Proof. SKIP FOR NOW $\square$

Exercise 28. Let F be increasing and right continuous, and let μ_F be the associated measure. Then $\mu_F(\{a\}) = F(a) - F(a-)$, $\mu_F([a, b]) = F(b) - F(a-)$, $\mu_F((a, b]) = F(b) - F(a)$, and $\mu_F((a, b)) = F(b-) - F(a)$.

Proof. First, consider an increasing sequence $\{a_n\}_1^\infty$ such that each $a_n < a$ and $\lim_{n \rightarrow \infty} a_n = a$. Then

$$\begin{aligned} \mu_F(\{a\}) &= \mu_F\left(\bigcap_1^\infty (a_n, a]\right) \\ &= \lim_{n \rightarrow \infty} \mu_F((a_n, a]) \\ &= \lim_{n \rightarrow \infty} F(a) - F(a_n) \\ &= F(a) - F(a-). \end{aligned}$$

Next we consider (a, b) . Let $\{b_n\}_1^\infty$ be an increasing sequence whose limit is b where each $b_n < b$. Then

$$\begin{aligned} \mu_F((a, b)) &= \mu_F\left(\bigcup_1^\infty (a, b_n]\right) \\ &= \lim_{n \rightarrow \infty} \mu_F(a, b_n] \\ &= \lim_{n \rightarrow \infty} F(b_n) - F(a) \\ &= F(b-) - F(a). \end{aligned}$$

The other two equalities follow immediately from these two, since

$$\begin{aligned}
\mu_F([a, b)) &= \mu_F(\{a\} \cup (a, b)) \\
&= \mu_F(\{a\}) + \mu_F((a, b)) \\
&= F(a) - F(a-) + F(b-) - F(a) \\
&= F(b-) - F(a-),
\end{aligned}$$

and by similar logic,

$$\begin{aligned}
\mu_F([a, b]) &= \mu_F([a, b)) + \mu_F(\{b\}) \\
&= F(b-) - F(a-) + F(b) - F(b-) \\
&= F(b) - F(a-).
\end{aligned}$$

□

Exercise 29. Let E be a Lebesgue measurable set.

- a. If $E \subset N$ where N is the nonmeasurable set described in §1.1, then $m(E) = 0$.
- b. If $m(E) > 0$, then E contains a nonmeasurable set. (It suffices to assume $E \subset [0, 1]$. In the notation of §1.1, $E = \bigcup_{r \in R} E \cap N_r$.)

Proof.

- a. Given $r \in R$, define E_r to be

$$E_r = \{x + r : x \in E \cap [0, 1 - r)\} \cup \{x + r - 1 : x \in E \cap [1 - r, 1)\}.$$

Immediately, we can see that

$$\begin{aligned}
m(E_r) &= m(\{x + r : x \in E \cap [0, 1 - r)\} \cup \{x + r - 1 : x \in E \cap [1 - r, 1)\}) \\
&= m(\{x + r : x \in E \cap [0, 1 - r)\}) + m(\{x + r - 1 : x \in E \cap [1 - r, 1)\}) \\
&= m(\{x : x \in E \cap [0, 1 - r)\}) + m(\{x : x \in E \cap [1 - r, 1)\}) \\
&= m(\{x : x \in E \cap [0, 1 - r)\} \cup \{x : x \in E \cap [1 - r, 1)\}) \\
&= m(E),
\end{aligned}$$

where the third equality comes by the translation invariance of m . Using a similar argument to that of §1.1, we see that

$$m\left(\bigcup_{r \in R} E_r\right) \leq m([0, 1)) = 1,$$

but

$$m\left(\bigcup_{r \in R} E_r\right) = \sum_{r \in R} m(E_r) = \sum_{r \in R} m(E).$$

Clearly, the final expression above cannot be finite unless $m(E) = 0$, so since it is forced to be ≤ 1 (i.e. finite), in fact $m(E) = 0$.

b. Since $X = \bigcup_{z \in \mathbb{Z}} I_z$ where each $I_z = [z, z + 1)$,

$$0 < m(E) = m\left(\bigcup_{z \in \mathbb{Z}} I_z \cap E\right) = \sum_{z \in \mathbb{Z}} m(I_z \cap E),$$

so $m(I_j \cap E) > 0$ for some $j \in \mathbb{Z}$. Then

$$(I_j \cap E) - j \subset I_0$$

and

$$m(I_j \cap E) - j = m(I_j \cap E) > 0,$$

so since $I_0 = \bigcup_{r \in R} N_r$, there exists a k such that

$$m(((I_j \cap E) - j) \cap N_k) > 0$$

Denote $((I_j \cap E) - j) \cap N_k$ by F . Note that part **a.** holds for each N_r in §1.1 since the choice of N was arbitrary and each N_r consists of exactly one representative from each equivalence class. It follows from **a.**, then, that since $F \subset N_k$ and $m(F) > 0$, F is not Lebesgue measurable.

Finally, notice that $F + j \subset I_j \cap E \subset E$, so $F + j$ is a nonmeasurable set contained in E since if $F + j$ were measurable, F would be as well.

□

Exercise 30. If $E \in \mathcal{L}$ and $m(E) > 0$, for any $\alpha < 1$ there is an open interval I such that $m(E \cap I) > \alpha m(I)$.

Proof. UNFINISHED

□